Development of an IoT-Based Smart Waste System with RTC Time Logging for Cleaning Assistance

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***Abstract*— Maintaining cleanliness in public spaces is a constant challenge, especially when trash bins aren't monitored regularly. Overflowing waste can create unsanitary conditions and public discomfort. To address this issue, the authors developed an IoT-based Smart Waste System that uses the Blynk app for real-time monitoring and alerts, along with a Real-Time Clock (RTC) module to record events. The system tracks the bin's fill level using an ultrasonic sensor and records the times when the bin becomes full or is emptied. It also includes a basic prediction feature to estimate when the bin will likely fill up again. Based on the results of research that has been carried out, there are sensors with good accuracy, such as ultrasonic sensors with an RMSE value of approximately 0.062 cm, and RTC modules with an RMSE value of approximately 75 seconds. However, the prediction system, although functional, has an RMSE value of around 7.7 hours, which shows that further improvement is needed. Overall, the system performed well under expected conditions and offers a more efficient alternative to traditional waste collection methods that rely on manual inspections.**

Keywords—Smart Bin, IoT, Waste Management, Ultrasonic Sensor, Real-Time Clock

I. Introduction

As part of daily lives, people naturally desire a clean and pleasant environment [1]. However, this desire is not always reflected in their actions; a simple example is that there is still a lot of garbage scattered, trash bins that are piled up, and public places that look slum due to the lack of individual awareness and sense of responsibility [2], [3]. This situation becomes a serious concern for both the community and cleaning administrators, especially in urban environments where population density significantly increases waste generation rates, reaching up to 2.01 billion tonnes globally in 2016 and projected to grow to 3.4 billion tonnes by 2050 [4], [5]. Even though cleaning agencies have placed waste bins in strategic public areas, these bins are often not used effectively. Because without monitoring, trash may accumulate until it overflows. This creates unpleasant odors and unsanitary conditions. In extreme cases, unattended garbage can attract pests and pose health hazards to nearby communities. Studies have shown that areas with unmanaged waste experience higher rates of vector-borne diseases and microbial contamination, especially in tropical urban zones [6].

In order to have a cleaner and healthier public environment, it's important to have a system that helps cleaning staff manage waste more efficient. One practical solution is to equip trash bins with sensors and notification systems to help detect when a bin is full and notify cleaning staff accordingly [7], [8]. With this help, overflow and unnecessary inspections could be minimized. Adding an RTC module to the smart waste bin system can increase the probability of the system's ability to collect behavioral data by recording the time when the bin has reached its full capacity and when it was last cleaned. After the data has been collected, users can use it to analyze the cleaning periods and determine locations with the highest disposal activity. Based on this information, cleaning schedules can be adjusted with more accuracy, and resources can be used more efficiently [9].

In this study, the authors proposed a solution to the problem of waste management: an Internet of Things (IoT)-based smart waste system that comes with a real-time clock (RTC). This system can record time and monitor waste levels. Additionally, the system can predict when waste bin will reach its full capacity. The methodology that is used for predicting waste levels involves analyzing filling events with timestamps. The system aims to reduce manual checks, prevent waste buildup, and support better waste management plans with real-time data.

II. Theory

A. Traditional Waste Management and Its Issues

In many cities, garbage collection still follows rigid schedules, regardless if the bins are full or not [4]. This can be a problem since sometimes bins are full long before pickup day, which may lead to unpleasant smells and attracting rodents or insects [10]. In the other cases, bins can still be half-empty when the garbage truck arrives to clean, which is not very effective. These inefficiencies will not just affect the overall cleanliness of public areas but also strain budgets by increasing labor and fuel usage [4]. Therefore, without real-time insights into bin conditions, it's hard for cleaning teams to choose and decide which areas need attention first or to make smart adjustments to their collection routes.

B. Integration of IoT Technology in Waste Management

The integration of Internet of Things (IoT) technologies in waste management makes it possible to have real-time monitoring and automation [11]. For example, a sensor-based system that can check fill levels and transmit the data wirelessly. Specifically in this study, the authors use the HC-SR04 ultrasonic sensor to measure how far the trash has risen in the bin [12]. An ESP32 microcontroller processes this data and sends it to the Blynk platform with the help of Wi-Fi. This setup helps cleaning staff to monitor bins in a flexible way.

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For larger areas or low-power applications, technologies like LoRaWAN could be a better option due to their wide range and energy efficiency [13], [14].

C. Role of RTC Module in Time Analysis

The authors integrated the RTC module, specifically the DS3231, to improve the usability for the system. The module is used in various IoT projects that require accurate time tracking, such as smart irrigation systems and environmental recording systems [15]. Tracking the time will allow the system to record when the sensor detects the trash level and the last time the trash was emptied accurately. This will help the system to analyze waste collection patterns, calculate estimated usage times, and generate predictive analysis to provide an accurate schedule, which helps to solve the problem of overflowing trash bins [9].

D. Related Smart Waste System Studies

Several earlier projects have explored similar ideas: combining environmental monitoring with IoT technologies. However, many of these systems either lacked time-based data logging or used only static detection thresholds. In a study conducted by Yusof et al., a system for basic content level notification was created using ultrasonic sensors, while in another study conducted by Ismail et al., a smart trash can system with mobile notifications was proposed [7].

Other researchers, like Bhuvaneswari et al. , build a smart garbage monitoring system that integrates distance measurement and the ThingSpeak platform to monitor waste level with helpful data using basic components like the HC-SR04 and ESP8266 [16]. Nur Fatimah also shared a working example of a Blynk-connected smart bin primarily for educational purposes and proof of concept [17]. There are also commercial systems, like Sensoneo's fill-level monitoring system, that offer advanced monitoring for cities [8]. But these are often designed for larger operations and may not be cost-effective for smaller-scale urban setups.

What sets our project apart is the addition of reliable time logging. The presence of RTC logging in the Sensing & Time Layer provides significant advantages in data recording accuracy. With the RTC module, each garbage height measurement sent by the ultrasonic sensor will be accompanied by a precise timestamp, enabling historical analysis of garbage disposal trends. This is not only for real-time monitoring facilities but also as a support for data-driven decision-making to make waste scheduling more efficient. The automatic integration of notifications using the Blynk application at the application and server layers aims to ensure that managers receive notifications immediately when the ESP32 detects that the trash bin capacity has exceeded the threshold. Compared to conventional methods that rely on manual inspection, this system minimizes response delays and reduces the risk of overflowing trash bins. Combining RTC logging and instant notifications offers higher efficiency and reliability.

III. Methods

A. Research Methodology

The authors of this project used IoT-based development, which is ideal for real-time detection, remote notification, and system automation. This methodology provides a structured approach to ensure that every component in the system functions effectively and efficiently. The design accommodates the dynamic nature of IoT systems, where data is collected and acted upon immediately. This methodology consists of five interconnected phases: initialization, detection, monitoring, notification, and user response. Each phase plays a crucial role in the overall workflow. This begins with system setup and sensor activation, followed by event or condition identification, continuous data tracking, notification delivery, and processing user interactions.

The hardware, as well as the HC-SR04 ultrasonic sensor and ESP32 microcontroller, is configured and connected to the Blynk IoT platform during the initialization process. The detection phase begins once waste enters the bin; the ultrasonic sensor calculates the distance to the waste surface to assess the fill level. The monitoring phase involves continuous data collection. The threshold value will be reached when it is indicated that the bin is full, and the system will proceed to the notification phase. The notification phase is when the alert is sent to the Blynk app in real time. During the user response phase, the user will empty the bin, which will reset the system and start a new cycle back to the beginning.

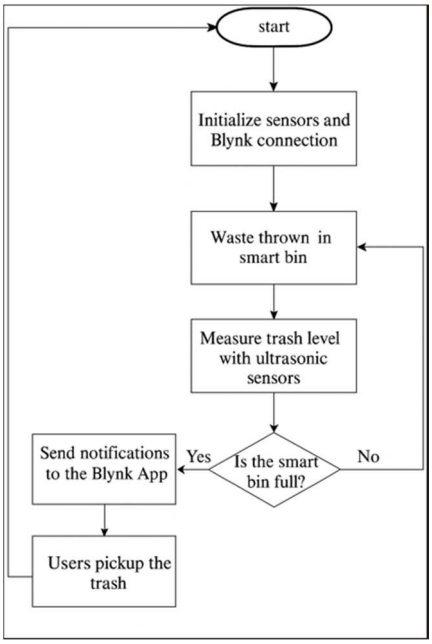

Fig. 1. System flow design

Fig. 1 shows a flowchart of the smart trash can. The system initializes the sensor and connects it to Blynk. Then the process will begin when trash is thrown into the trash bin, and then the ultrasonic sensor measures the distance to determine whether the bin is full. If the bin reaches its maximum capacity, the user will receive a notification and will be asked to empty the trash can.

B. System Design

This system is providing easy waste monitoring with three main components that are ESP32 microcontroller, a Real-Time Clock (RTC) module, and an ultrasonic sensor. The ultrasonic sensor detects the distance to the surface of the waste that allowed the system to calculate the waste level inside the bin. The RTC module has an accurate time recording that allowing the system to record the time when the bin reaches maximum capacity. Meanwhile, the ESP32 microcontroller acts as the central processing and communication unit, overseeing all data processing operations and managing connections to external services.

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The ESP32 operates using an ultrasonic sensor to measure the distance. Then, the microcontroller evaluates the reading based on the defined threshold. This evaluation determines whether the bin has reached its maximum capacity. The ESP32 will activate the RTC module to record the timestamp if the sensor detects that the waste level exceeds the threshold. Then a notification will be sent via Wi-Fi to the Blynk platform. Within Blynk, users or sanitation staff are able to track the current condition of the waste bin, get alerts upon reaching fullness, and take prompt measures based on the information received.

For power supply, the entire system is powered by a portable 5V power bank connected via a USB interface. This portable power source enhances the flexibility of deployment, allowing the Smart Waste Bin to be used in various locations without dependency on fixed electrical infrastructure. It is important to note that the current design does not include any actuators or mechanical systems for waste disposal. Instead, the focus of this project lies solely on sensing the bin's status, recording relevant data such as time of fullness, and sending notifications for timely response.

Fig. 2 shows the system architecture of a smart waste bin, including the layered design. The system is powered by a power bank connected to the main controller, an ESP32 microcontroller. The ESP32 manages data from an ultrasonic sensor (HC-SR04) to detect the waste level inside the bin and receives real-time clock data from the RTC DS3231 module to timestamp actions. All data is processed and sent to the Blynk application via Wi-Fi, allowing users to monitor the waste level in real time and receive notifications when the bin is full.

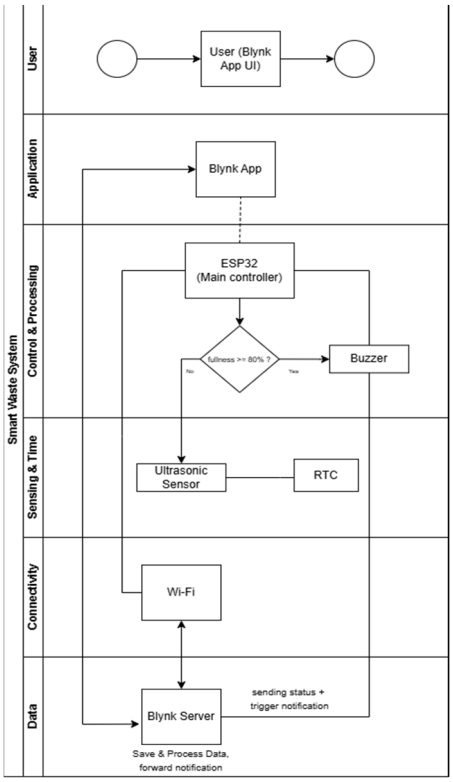

Fig. 2. Block diagram

Fig. 3 Blynk Interface shows a visual representation of the Blynk application as used in the Smart Waste System project. It displays elements that would be seen by a user or cleaning staff member monitoring the waste bin. It includes useful information such as the current fill level of the bin determined by the ultrasonic sensor, a duration prediction of when the bin will be full, and alerts when the bin reaches a full state. The interface also provides timestamps of when the bin was last l, as well as a historical graph of the duration of the bin until it is full. In essence, the Blynk interface provides a real-time, user-friendly way to track the status and activity of the smart waste bin remotely.

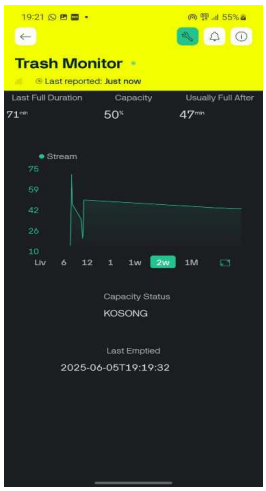

Fig. 3. Blynk interface

C. Data Validation and Accuracy Testing

There are two testing methods carried out to measure the accuracy of this Smart Waste Management system, the testing methods are black box testing and RMSE, namely root mean square error.

In the black box testing method, testing is done by making a comparison for the HC-SR04 ultrasonic sensor readings used in the system with measurements using a ruler manually. In this system design, the HC-SR04 ultrasonic sensor is positioned vertically above the trash can to ensure that the acoustic pulse travels perpendicularly to the waste surface. This orientation helps minimize angular reflection and measurement error. Lab-scale validations of similar low-cost ultrasonic sensors have shown that consistent, orthogonal mounting under controlled lighting and environmental noise achieves repeatable accuracy, with deviations below 1 %, even under varying ambient light and background interference [18].

The distance to the surface of the garbage can is measured manually using a ruler, then compared with the sensor data in the BLYNK application. In addition, for the RTC DS3231 module contained in the system, it is manually tested by measuring the similarity or compatibility of the Time recording results on the system with a digital clock as a reference for actual time. After testing with the black box method is finished, the sensor reading data is compared with the actual measurement results using the RMSE formula. If the calculation of the value obtained from the RMSE is smaller, the higher the accuracy of the sensor in the system.

D. Hardware List

The hardware list needed to design this smart waste management bin system are ESP32 development board, HC-SR04 ultrasonic sensor, RTC module DS3231, jumper wires, breadboard, and 5V power source.

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IV. Results

A. Black Box Testing

To check how well the Smart Waste System works, the authors did some simple tests. The authors wanted to see if the main parts, for example the sensors and timers, were doing their jobs without looking at the complicated computer code inside to check if the important pieces were working as they should.

TABLE I. Black Box Testing Result for Ultrasonic Sensor (Distance to Trash Level)

Trial	Actual Distance (cm)	Sensor Reading (cm)	Error (cm)
1	5	5.06	0.06
2	10	9.9	0.1
3	20	20.02	0.02
4	22	22.04	0.04

[a.] Black Box Testing Result for Ultrasonic Sensor (Distance to Trash Level)

Table I shows the result of testing the HC-SR04 ultrasonic sensor's accuracy. It compares the actual distance from the sensor to a surface (representing the trash level) measured manually in centimeters, against the distance reading reported by the sensor. The "Deviation" column shows the difference between the actual and sensor readings. This table helps evaluate how precisely the ultrasonic sensor measures distance, which directly impacts the system's ability to detect the bin's fill level.

Table I shows that measurements from the HC-SR04 ultrasonic sensor produce small differences between the actual distance and the distance read by the sensor, with errors ranging from approximately 0.02 cm to 0.10 cm. These low deviation values define the high level of precision and consistency in detecting waste in this system. The RMSE value of 0.062 cm demonstrates that sensor readings remain stable despite potential disturbances from real-world conditions such as weather changes or environmental noise. Compared to previous studies [7], [9], which did not provide detailed accuracy metrics and thus could not reduce noise levels, this system achieves significantly more precise measurement accuracy. This accuracy is crucial because even the smallest error can impact the efficiency of waste management operations.

TABLE II. Black Box Testing Result for RTC Module (Time when bin was last emptied)

Trial	Real Time	RTC Output	Time Difference
1	22-05-2025 14:22:31	22-05-2025 14:23:48	77 seconds
2	22-05-2025 15:01:15	22-05-2025 15:02:29	74 seconds
3	23-05-2025 09:22:37	23-05-2025 09:23:52	75 seconds
4	23-05-2025 11:53:11	23-05-2025 11:54:25	74 seconds

[b.] Black Box Testing Result for RTC Module (Time when bin was last emptied)

Table II shows the results of testing the accuracy of the DS3231 RTC module. These results compare the real time recorded from an external clock recorded by the system with the possible time when the trash bin is empty. The "Time Difference" column indicates the discrepancy between the real time and the RTC output in seconds. This data is critical for evaluating how well the RTC module tracks time, which is essential for the system's timestamping and prediction features.

The results shown in Table II indicate that the DS3231 Real-Time Clock (RTC) module produces a consistent time difference of between 74 and 77 seconds compared to the external reference time. The RMSE value of approximately 75 seconds indicates sufficient stability in time recording for waste level monitoring applications, although the accuracy shown has not yet reached the level of second-to-second synchronization. Consistency is crucial to ensure reliable time recording for full and empty bins, enabling analysis of waste disposal behavior and prediction of emptying schedules. Unlike previous studies that did not use an RTC or evaluate its accuracy, this system offers an advantage with its measurable time logging capability for predictive analysis.

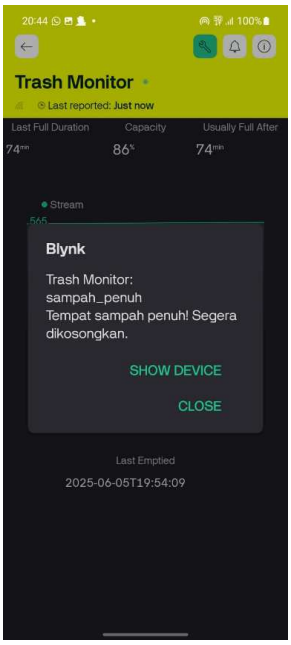

Fig. 4. Full notification

Fig. 4 shows the BLYNK successfully notified user when the capacity was above the threshold. This confirmes the notification feature's functionality.

B. RTC-Based Full-Time Prediction Testing

To understand how well the system can predict when a bin will be full, the authors looked at the last 10 times the bin reached full capacity. The prediction was made by calculating the average time between those full events. This simple method gives a basic estimate for the next full time, which can be helpful but is not always accurate, especially in changing or unpredictable environments. More advanced time-series forecasting techniques such as Autoregressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM) models have been successfully applied in similar IoT-based prediction tasks, showing improved accuracy and adaptability in dynamic environments [19], [20]

TABLE III. Prediction vs Actual Based on RTC

Trial	Full Timestamp	Actual Full (hours)	Predicted Full (hours)	Time Difference
1	24-05-2025 13:12:51	-	-	-
2	24-05-2025 15:23:14	2	-	-

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Trial	Full Timestamp	Actual Full (hours)	Predicted Full (hours)	Time Difference
3	25-05-2025 12:20:35	20	2	18
4	25-05-2025 16:44:46	4	11	7
5	26-05-2025 09:25:34	16	12	4
6	26-05-2025 19:19:27	9	10	1
7	27-05-2025 11:24:31	18	10	8
8	27-05-2025 21:34:53	10	10	0
9	28-05-2025 12:54:21	15	11	4
10	28-05-2025 19:57:02	9	11	2

c. Prediction vs Actual Based on RTC

Table III shows the test results of the system for predicting when the trash bin will be full based on the data from the RTC. This table shows four aspects of the ten test cases, namely "Full Time" when the trash bin is detected as full. There is also the actual full time and the predicted full time which is calculated by the system. There is also a "Time Difference" column showing the difference between the actual full time and the predicted full time. This table is important for achieving the system's effectiveness in predicting the algorithm to optimize waste management for users.

The results in Table III show the system's ability to predict full time based on data collected from the RTC module. The difference between actual full time and system predictions ranges from 0 to 18 hours, with an RMSE value of approximately 7.7 hours. This accuracy is sufficient to provide an estimate of the emptying schedule, but there is also the possibility of significant deviations under unstable disposal patterns. The prediction method used is the average interval between full events, which is simple and sensitive to changes in user behavior. Compared to previous studies, which generally lack prediction features, this system has the advantage of providing direct estimates of full time through low-cost IoT devices. However, this prediction method can still be improved to achieve better accuracy, especially in environments with high variability.

C. *Accuracy Evaluation Using Root Mean Square Error (RMSE)*

To evaluate the overall accuracy of the system, RMSE was calculated for three components: the ultrasonic sensor, the RTC module, and the predicted full-time estimation. With formula as below:

$$RMSE = \sqrt{\left(\frac{(\sum(P[i]-A[i])^2)}{n}\right)} \quad (1)$$

Where:

- P[i] = Sensor's result
- A[i] = Actual measurement
- i = index of iteration
- n = Amount of iteration

1) *Ultrasonic Sensor*

Using actual vs. sensor distances (errors: 0.06, 0.1, 0.02, 0.04 cm), the RMSE was around 0.062 cm. This low value means that the sensor provides high-accuracy readings even under unstable light and background interference conditions.

2) *RTC Module*

Comparing system timestamps with a reference clock (errors: 77, 74, 75, 74 seconds), the RMSE was around 75 seconds. These stable differences are enough for tracking bin full events and supporting time-based analysis.

3) *Prediction System*

The prediction system used average intervals between full-bin events. Based on 10 comparisons (errors: 18, 7, 4, 1, 8, 0, 4, 2 hours), the RMSE was around 7.7 hours. While this is acceptable for basic estimation, the system could be improved with more data and advanced forecasting methods like ARIMA or LSTM [19]. Overall, the result means there is strong accuracy for trash level sensing and timestamp logging clear, with a room to improve for the predictive feature.

D. *Advantages & Limitations*

This Smart Waste System has several advantages. First, the integration of ultrasonic sensors provides high measurement accuracy (RMSE $\approx$ 0.062 cm) to ensure reliable detection of full trash bin levels, thereby minimizing the possibility of false notifications. Second, the use of an RTC module enables more accurate time recording (RMSE $\approx$ 75 seconds), allowing optimal provision of waste disposal behavior data for scheduling emptying. Third, the use of low-cost components and a simple architecture makes this system affordable and adaptable to various urban environments. Fourth, real-time notification features via the Blynk platform enable waste management personnel to respond quickly, thereby reducing the risk of overflowing trash bins.

However, this system also has limitations on the prediction method used today. Based on the full interval average, the RMSE value results in about 7.7 hours. This value is sufficient for basic schedule estimation, but less accurate in environments with irregular discharge patterns. In addition, the system's reliance on a stable Wi-Fi connection is also a limitation so that its application will be limited to areas with inadequate networks.

V. Conclusion

This research addresses a common issue in public sanitation: trash bins that overflow because they aren't emptied on time. By using sensors, real-time notifications, and a clock module to log when bins reach capacity, the system helps cleaning staff respond more efficiently. Tests showed that the sensors work accurately and that the time-logging feature is stable and reliable. While the prediction feature is still basic, using average intervals between events, it lays the groundwork for smarter forecasting. The system demonstrates how low-cost IoT tools can streamline waste collection. With future upgrades, such as solar power or machine learning integration, it could scale effectively in larger, more complex settings. As cities adopt smarter infrastructure, tools like this could support cleaner streets and better-informed decisions through real-time data shared on municipal dashboards [11].

Author Contributions

Jeremy Axel Suryamiharja did the writing and investigation. Leony Agustin did the methodology and editing. Talitha Zasizki Gunara did the writing and methodology. Mochammad Haldi Widianto completed the scientific paper by considering revisions from reviewers until

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completion. Therefore, all authors contributed to this scientific article.

DATA AVAILABILITY

The data used for this study are openly available at: https://github.com/jejewepe/WasteManagementSystem

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